Analysis Subject: Nutrition

Extract Accurate (provide source) and Actionable information linking data to outcome (naïve example: to sleep 8h a day increase lifespan by 10 years on average based on source XXX).

An example of source is https://protocol.bryanjohnson.com/#bryans-protocol but it could be any scientifically accurate source such as WHO Guidelines, Scientific reviews, etc.

Types of Diets that Help Longevity

Mediterranean Diets

Main Nutrients
- Whole grains
- Fresh vegetables
- Fruits
- Beans
- Legumes
- Oils
- Vinegars
- Nuts
- Seeds
- Dairy
- Fish
- Eggs
- Fermented foods

Limitations
- Food sensitivities
- Severe allergies (especially to fish and nuts)
- Vegetarian/vegan
- Expensive
- Accessibility barriers
- Sensitive to high fiber intake

Mediterranean Diet is primarily related to 4 main cellular mechanisms to increase lifespan:

1. **Telomere Preservation**

Actionable Recommendation	Start eating a more mediterranean diet instead of the regular food you normally intake
Evidence	“Sticking to a Mediterranean diet may lead to a longer life. The findings are based on the study of telomeres.”

Longevity Impact	A mediterranean diet hinders the shortening of telomeres which is related to a lot of aging-related diseases, keeping a low risk for many diseases and increasing lifespan
Source	Brigham and Women's Hospital

Actionable Recommendation	Start eating a more plant rich foods in the mediterranean diet such as veggies, fruits, legumes, beans, and whole grains
Evidence	"Previous studies have shown that plant-rich dietary patterns, such as the Mediterranean diet, are associated with longer telomeres."
Longevity Impact	A mediterranean diet hinders the shortening of telomeres which is related to a lot of aging-related diseases, keeping a low risk for many diseases and increasing lifespan
Source	NIH PubMed Central

Actionable Recommendation	Start to have a high intake of vegetables, fruits, nuts, legumes, and grains (mainly unrefined); a high consumption of olive oil but a low intake of saturated fat; a moderately high intake of fish; a low intake of dairy products, meat, and processed meat; and a regular but moderate intake of alcohol (specifically wine with meals).
Evidence	"The Mediterranean diet (MedDiet) is considered to be 1 of the most recognized diets for disease prevention and healthy aging, partially due to its demonstrated anti-inflammatory and antioxidative properties which may impact on telomere length (TL)."
Longevity Impact	A mediterranean diet hinders the shortening of telomeres which is related to a lot of aging-related diseases, keeping a low risk for many diseases and increasing lifespan
Source	NIH PubMed Central

Actionable Recommendation	Start to have a high intake of vegetables, fruits, nuts, legumes, and unrefined grains for a longer telomere length
Evidence	"Mediterranean diet would be correlated with longer telomere length."

Longevity Impact	A mediterranean diet hinders the shortening of telomeres which is related to a lot of aging-related diseases, keeping a low risk for many diseases and increasing lifespan
Source	University of New Mexico

Actionable Recommendation	Start eating more colorful fruits, vegetables, healthy fats, legumes, and whole grains to help preserve telomere length
Evidence	"The Mediterranean diet may help protect the length of telomeres, which sit at the end of chromosomes and protect the ends from fraying."
Longevity Impact	A mediterranean diet hinders the shortening of telomeres which is related to a lot of aging-related diseases, keeping a low risk for many diseases and increasing lifespan
Source	Harvard

Actionable Recommendation	Eat more extra virgin olive oil, fatty fish, leafy greens, berries, and nuts for telomere support and growth.
Evidence	"The Mediterranean diet is beneficial for maintaining telomere dynamics and alleviating age-related illnesses."
Longevity Impact	A mediterranean diet hinders the shortening of telomeres which is related to a lot of aging-related diseases, keeping a low risk for many diseases and increasing lifespan
Source	NIH PubMed Central

Actionable Recommendation	Start eating a more mediterranean diet instead of the regular food you normally intake
Evidence	**"Women who more closely followed a Mediterranean-style diet had longer telomeres."**
Longevity Impact	A mediterranean diet hinders the shortening of telomeres which is related to a lot of aging-related diseases, keeping a low risk for many diseases and increasing lifespan
Source	"Adherence to a Mediterranean diet linked to longer telomere length" by Nilay Shah and Ravi Shah

2. **Protect membranes from damage**

Actionable Recommendation	Start adding more olive oil and nuts aspect of mediterranean diets to your daily intake of food which increases your lipid intake and in turn, protects membranes from damage
Evidence	“Fat from olive oil and nuts boosts the numbers of two key cellular structures [lipid droplets] and protects membranes from damage.”
Longevity Impact	A mediterranean diet is based largely on olive oil and nuts, so as it protects membranes from damage, cellular health increases, and so does lifespan.
Source	Stanford Medicine

Actionable Recommendation	Start eating more virgin olive oil or nuts as part of the mediterranean diet as it helps support membranes in your body
Evidence	“The Mediterranean-style diet affects the lipid metabolism that is altered in hypertensive (blood pushing against artery walls is consistently too high) patients, influencing the structural membrane properties. The erythrocyte membrane modulation described provides insight in the structural bases underlying the beneficial effect of a Mediterranean-style diet in hypertensive subjects.”
Longevity Impact	A mediterranean diet is based largely on olive oil and nuts, so as it protects membranes from damage, cellular health increases, and so does lifespan.
Source	NIH PubMed Central

Actionable Recommendation	Incorporate more extra virgin oil and nuts into your everyday meals to help support the membranes in your body
Evidence	“The effect of a Mediterranean-style diet supplemented with nuts or virgin olive oil on erythrocyte membrane properties in 36 hypertensive participants after 1 year of intervention. Erythrocyte membrane lipid composition, structural properties of reconstituted erythrocyte membranes, and serum concentrations of inflammatory markers are reported. After the

	intervention, the membrane cholesterol content decreased, whereas that of phospholipids increased in all of the dietary groups; the diminishing cholesterol:phospholipid ratio could be associated with an increase in the membrane fluidity. Moreover, reconstituted membranes from the nuts and virgin olive oil groups showed a higher propensity to form a nonlamellar inverted hexagonal phase structure that was related to an increase in phosphatidylethanolamine lipid class.”
Longevity Impact	A mediterranean diet is based largely on olive oil and nuts, so as it protects membranes from damage, cellular health increases, and so does lifespan.
Source	American Heart Association Journals

Actionable Recommendation	Eat more fruits, vegetables and salad, bread and whole grain cereals, potatoes, legumes/beans, nuts, and seeds to support the membranes your body needs
Evidence	“The Mediterranean diet improves the systemic lipid and DNA oxidative damage in metabolic syndrome individuals.”
Longevity Impact	A mediterranean diet is based largely on olive oil and nuts, so as it protects membranes from damage, cellular health increases, and so does lifespan.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more olive oil and fish products as it supports membranes greatly
Evidence	“High adherence to the MedDiet during adolescence is positively associated with omega-3 FAs in RBC membranes.”
Longevity Impact	A mediterranean diet is based largely on olive oil and nuts, so as it protects membranes from damage, cellular health increases, and so does lifespan.
Source	NIH PubMed Central

Actionable Recommendation	Start having a high intake of fish, nuts and olive oil, which are natural sources of omega-3 (n-3) and n-9 fatty acids (FA), and

	a moderate intake of red meat and processed food to help support membranes in your body
Evidence	“Change to the Mediterranean diet results in particular changes in fatty acid patterns in the erythrocyte membrane.”
Longevity Impact	A mediterranean diet is based largely on olive oil and nuts, so as it protects membranes from damage, cellular health increases, and so does lifespan.
Source	“Fatty acid profiles in erythrocyte membranes following the Mediterranean diet – data from a multicenter lifestyle intervention study in women with hereditary breast cancer (LIBRE)” by Benjamin Seethaler et. al

3. **Support Mitochondrial Health**

Actionable Recommendation	Eat more mediterranean diet oriented foods, especially those with high essential oils and fiber support mitochondrial health the most
Evidence	“Cumulative evidence from pre-clinical studies indicates that the MD is rich in polyphenols, essential oils, and fiber and plays a beneficial role by stimulating mitochondrial biogenesis and exerting an antioxidant effect.”
Longevity Impact	A mediterranean diet helps stimulate and support mitochondrial health, which causes it to produce more ATP to repair tissues and reduce age related inflammation, leading to a longer life.
Source	NIH PubMed Central

Actionable Recommendation	Eat more extra virgin olive oil, fatty fish, berries, legumes, and whole grains, vegetables, nuts, and seeds to boost mitochondria microproteins and in turn efficiency all together
Evidence	“The researchers propose that the Mediterranean diet may work through a dual mechanism: directly lowering oxidative stress while also boosting mitochondrial microproteins.”
Longevity Impact	A mediterranean diet helps stimulate and support mitochondrial health, which causes it to produce more ATP to

	repair tissues and reduce age related inflammation, leading to a longer life.
Source	USC Leonard Davis

Actionable Recommendation	Eat more extra virgin olive oil, fatty fish, berries, legumes, and whole grains to boost mitochondria efficiency
Evidence	“Mediterranean diet (MedDiet) seems to affect the function of mitochondria further justifying the role of this diet in lowering the risk of negative health outcomes.”
Longevity Impact	A mediterranean diet helps stimulate and support mitochondrial health, which causes it to produce more ATP to repair tissues and reduce age related inflammation, leading to a longer life.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more whole grains, legumes, fruits, and vegetables as the primary sources of carbohydrate, protein and fiber; nuts and extra virgin olive oil, which provide the main dietary fats; and moderate amounts of fish, poultry, and dairy products, with limited red meat consumption as part of the mediterranean diet
Evidence	“The Med-Diet and its specific components may positively influence mitochondrial function.”
Longevity Impact	A mediterranean diet helps stimulate and support mitochondrial health, which causes it to produce more ATP to repair tissues and reduce age related inflammation, leading to a longer life.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more whole grains, legumes, fruits, and vegetables as the primary sources of carbohydrate, protein and fiber; nuts and extra virgin olive oil, which provide the main dietary fats; and moderate amounts of fish, poultry, and dairy products, with limited red meat consumption as part of the mediterranean diet

Evidence	"The Med-Diet and its specific components may positively influence mitochondrial function."
Longevity Impact	A mediterranean diet helps stimulate and support mitochondrial health, which causes it to produce more ATP to repair tissues and reduce age related inflammation, leading to a longer life.
Source	NIH PubMed Central

Actionable Recommendation	Eat more extra virgin olive oil, oregano, fatty fish, walnuts, and flaxseeds to support mitochondrial health.
Evidence	"Studies continue to report that Mediterranean-style diets that prioritize plants, healthy fats, and whole grains reduce mitochondrial damage."
Longevity Impact	A mediterranean diet helps stimulate and support mitochondrial health, which causes it to produce more ATP to repair tissues and reduce age related inflammation, leading to a longer life.
Source	The Institute for Functional Medicine

Actionable Recommendation	Eat more extra virgin olive oil, fatty fish, legumes, nuts, seeds, fruits, vegetables, and whole grains to support the mitochondria, especially the tiny proteins in it.
Evidence	"Mediterranean Diet May Boost Mitochondrial Signals."
Longevity Impact	A Mediterranean diet increases gut microbiota diversity and modulates or balances it, which helps prevent onset of chronic non-communicable degenerative diseases which can lower lifespan.
Source	USC Neuroscience Graduate Program

4. **Increase gut microbiota diversity and balance**

Actionable Recommendation	Start eating more fermented food and more fiber which increase gut microbiota diversity the most
Evidence	"MD is able to modulate the gut microbiota, increasing its diversity."

Longevity Impact	A Mediterranean diet increases gut microbiota diversity and modulates or balances it, which helps prevent onset of chronic non-communicable degenerative diseases which can lower lifespan.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more legumes, grains, vegetables, and fruits to improve your gut microbiota
Evidence	“The MD is also associated with characteristic modifications to gut microbiota, mediated through its constituent parts.”
Longevity Impact	A Mediterranean diet increases gut microbiota diversity and modulates or balances it, which helps prevent onset of chronic non-communicable degenerative diseases which can lower lifespan.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more olive oil, fish, and fiber to balance your gut microbiota better
Evidence	“A new Tulane University study suggests the Mediterranean diet's brain-boosting benefits may work by changing the balance of bacteria in the gut.”
Longevity Impact	A Mediterranean diet increases gut microbiota diversity and modulates or balances it, which helps prevent onset of chronic non-communicable degenerative diseases which can lower lifespan.
Source	Tulane University

Actionable Recommendation	Start eating more beans, peas, lentils, oats, and vegetables for more fiber for your gut microbiota
Evidence	“Mediterranean style diet has beneficial influence on the gut microbiota, so it may be considered for use in clinical management and prevention in diabesity.”
Longevity Impact	A Mediterranean diet increases gut microbiota diversity and modulates or balances it, which helps prevent onset of chronic

	non-communicable degenerative diseases which can lower lifespan.
Source	[Kansas State University]

Actionable Recommendation	Start eating more veggies, fruits, whole grains, and extra virgin olive oil for your gut microbiota
Evidence	"The Mediterranean Diet has many benefits, including…Supporting a healthy balance of gut microbiota (bacteria and other microorganisms) in your digestive system."
Longevity Impact	A Mediterranean diet increases gut microbiota diversity and modulates or balances it, which helps prevent onset of chronic non-communicable degenerative diseases which can lower lifespan.
Source	Cleveland Clinic

Actionable Recommendation	Try eating more fruits and vegetables to benefit your gut microbiota
Evidence	"The Mediterranean Diet (MedDiet) is abundant in ω3 fatty acids, fruits, and vegetables, possessing anti-inflammatory properties that contribute to the restoration of gut eubiosis." (gut eubiosis is the healthy, balanced state of you gut microbiota)
Longevity Impact	A Mediterranean diet increases gut microbiota diversity and modulates or balances it, which helps prevent onset of chronic non-communicable degenerative diseases which can lower lifespan.
Source	"Unraveling the intricate dance of the Mediterranean diet and gut microbiota in autoimmune resilience" by Christina Tsigalou et. al

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Plant Based and Plant Forward Diets

Main Nutrients

- Whole produce

- Legumes
- Grains
- Nuts and seeds
- Liquid fats (oils)
- NO MEAT

Limitations
- Calcium deficient
- Iron deficient
- Zinc deficient
- Vitamin B12 deficient
- Omega-3 deficient
- IBS condition
- Expensive

Plant-Based Diets are primarily related to 4 main cellular mechanisms to increase lifespan:

1. **Autophagy Activation**

Actionable Recommendation	Start eating more plant-derived foods, such as fruits, vegetables, and grains, as they help autophagy activation the most
Evidence	“Bioactive natural compounds derived from plants…have emerged as potent modulators of autophagy, offering a promising strategy to counteract aging and promote healthy lifespan.”
Longevity Impact	Autophagy is the body's internal recycling system to clear out damaged oragelles and misfolded proteins, so with more autophagy activation, the recycling system works harder and keeps the cells healthy and in turn increases the lifespan of the cell and ourselves in general. Polyphenols, found primarily in plants, also support and modulate autophagy.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more fruits, spices, teas, coffee and other plants for the compounds needed to support autophagy
Evidence	“Further experimental studies found out that fruits, spices, teas, coffee, and other plants, as well as their bioactive compounds, such as resveratrol, anthocyanin, curcumin, and

	tea polyphenols, could alleviate FLD by…regulating autophagy and ethanol metabolism.”
Longevity Impact	Autophagy is the body's internal recycling system to clear out damaged oragelles and misfolded proteins, so with more autophagy activation, the recycling system works harder and keeps the cells healthy and in turn increases the lifespan of the cell and ourselves in general. Polyphenols, found primarily in plants, also support and modulate autophagy.
Source	“Plant-Based Foods and Their Bioactive Compounds on Fatty Liver Disease: Effects, Mechanisms, and Clinical Application” by Hang-Yu Li et. al

Actionable Recommendation	Start eating more berries, flaxseeds, and dark chocolate for the most polyphenols for autophagy
Evidence	“Polyphenols are plant secondary metabolites and their ingestion may offer beneficial effects on biological processes, including cell proliferation, cell cycle, apoptosis and autophagy.”
Longevity Impact	Autophagy is the body's internal recycling system to clear out damaged oragelles and misfolded proteins, so with more autophagy activation, the recycling system works harder and keeps the cells healthy and in turn increases the lifespan of the cell and ourselves in general. Polyphenols, found primarily in plants, also support and modulate autophagy.
Source	“Regulation of autophagy by plant-based polyphenols: A critical review of current advances in glucolipid metabolic diseases and food industry applications” by Zhenyu Wang et. al

Actionable Recommendation	Start eating more common herbs and fruits, vegetables, herbs and spices to promote autophagy
Evidence	“Terpenoids from common herbs promote autophagy.”
Longevity Impact	Autophagy is the body's internal recycling system to clear out damaged oragelles and misfolded proteins, so with more autophagy activation, the recycling system works harder and keeps the cells healthy and in turn increases the lifespan of

	the cell and ourselves in general. Polyphenols, found primarily in plants, also support and modulate autophagy.
Source	Nature

Actionable Recommendation	Start eating more fruits, vegetables, whole grains, and legumes, cocoa, and tea to promote autophagy
Evidence	"One of the dietary ingredients common in plant-based diets is polyphenols"
Longevity Impact	Autophagy is the body's internal recycling system to clear out damaged oragelles and misfolded proteins, so with more autophagy activation, the recycling system works harder and keeps the cells healthy and in turn increases the lifespan of the cell and ourselves in general. Polyphenols, found primarily in plants, also support and modulate autophagy.
Source	NIH PubMed Central

Actionable Recommendation	Eat more of fruits, vegetables, and cereals to get more polyphenols and in relation, promote autophagy
Evidence	"Plant Polyphenols for Aging Health…Their Autophagy Modulating Properties in Age-Associated Diseases."
Longevity Impact	Autophagy is the body's internal recycling system to clear out damaged oragelles and misfolded proteins, so with more autophagy activation, the recycling system works harder and keeps the cells healthy and in turn increases the lifespan of the cell and ourselves in general. Polyphenols, found primarily in plants, also support and modulate autophagy.
Source	NIH PubMed Central

2. **AMPK Activation**

Actionable Recommendation	Eat more fruits like apples and berries, as vegetables like broccoli to mimic metformin and canagliflozin and activate AMPK
Evidence	"Natural/naturally derived plant compounds, metformin and canagliflozin, were shown to activate AMPK."

Longevity Impact	AMPK is an enzyme that regulates cellular energy and is needed to promote mitochondrial biogenesis to produce more ATP and improve metabolic efficiency. With plant-based diets and foods increasing AMPK activation, more cellular energy can be produced throughout the body, helping keep the body healthy and lead to a longer life.
Source	NIH PubMed Central

Actionable Recommendation	Eat more grapes, berries, peanuts, tree turmeric, and dark chocolate to get the most resveratrol and berberine which is needed for AMPK activation
Evidence	"Resveratrol, a natural phytochemical, increases glucose uptake in insulin-resistant 3T3-L1 adipocytes by increasing the phosphorylation of pAkt and downstream AMPK activation…Berberine is a naturally occurring plant-derived compound that can indirectly activate AMPK."
Longevity Impact	AMPK is an enzyme that regulates cellular energy and is needed to promote mitochondrial biogenesis to produce more ATP and improve metabolic efficiency. With plant-based diets and foods increasing AMPK activation, more cellular energy can be produced throughout the body, helping keep the body healthy and lead to a longer life.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more plant-based milk and yogurt to get more plant extracts for AMPK activation
Evidence	"Combinations of specific plant extracts with particular nutrients might more effectively activate AMPK signaling."
Longevity Impact	AMPK is an enzyme that regulates cellular energy and is needed to promote mitochondrial biogenesis to produce more ATP and improve metabolic efficiency. With plant-based diets and foods increasing AMPK activation, more cellular energy can be produced throughout the body, helping keep the body healthy and lead to a longer life.
Source	"Natural products targeting AMPK signaling pathway therapy, diabetes mellitus and its complications" by Min Li et. al

Actionable Recommendation	Start eating more
Evidence	“A particularly large class of AMPK activators are natural products of plants derived from traditional herbal medicines.”
Longevity Impact	AMPK is an enzyme that regulates cellular energy and is needed to promote mitochondrial biogenesis to produce more ATP and improve metabolic efficiency. With plant-based diets and foods increasing AMPK activation, more cellular energy can be produced throughout the body, helping keep the body healthy and lead to a longer life.
Source	“Regulation of AMP-activated protein kinase by natural and synthetic activators” by David Grahame Hardie

Actionable Recommendation	Start eating more broccoli sprouts, grapes, berries, turmeric, green tea, and flaxseeds to promote AMPK activation.
Evidence	“The mechanisms through which different dietary compounds, from plant foods, affect the AMPK pathway.”
Longevity Impact	AMPK is an enzyme that regulates cellular energy and is needed to promote mitochondrial biogenesis to produce more ATP and improve metabolic efficiency. With plant-based diets and foods increasing AMPK activation, more cellular energy can be produced throughout the body, helping keep the body healthy and lead to a longer life.
Source	Universidad San Francisco De Quito

Actionable Recommendation	Start eating more soy foods, fiber-rich plant foods, and turmeric to promote AMPK activation
Evidence	“This review delves into ten flavonoids identified as AMPK activators, including baicalein, dihydromyricetin, bavachin, 7-O-MA, derrone, and alpinumisoflavone.” (all plant-based compounds)
Longevity Impact	AMPK is an enzyme that regulates cellular energy and is needed to promote mitochondrial biogenesis to produce more ATP and improve metabolic efficiency. With plant-based diets and foods increasing AMPK activation, more cellular energy

	can be produced throughout the body, helping keep the body healthy and lead to a longer life.
Source	“Plant-Derived Flavonoids as AMPK Activators: Unveiling Their Potential in Type 2 Diabetes Management through Mechanistic Insights, Docking Studies, and Pharmacokinetics” by Dong Oh Moon

3. **mTOR Inhibition**

Actionable Recommendation	Eat more fruits and veggies like broccoli, green tea, soy, turmeric, grapes, onions, strawberries, blueberries, and mangoes.
Evidence	“Eating plants—and specifically cruciferous veggies—decreases mTOR activation *and* provides natural mTOR inhibition.”
Longevity Impact	With more mTOR inhibition cells stop growing and dividing as much, and rather, focus on maintaining and repairing to keep the cells and body healthy and ensure a healthy, long life. mTOR can also cause tumor growth but the inhibition can lower the cancer risk. mTOR is also caused primarily by leucine, so with low leucine intake, mTOR is inhibited more.
Source	Plant Based Life

Actionable Recommendation	Start eating more soy foods, legumes, and nuts for the high-leucine and the polyphenol needed for mTOR inhibition
Evidence	“Plant-based diets have all been linked to decreased risk for various cancers.”
Longevity Impact	With more mTOR inhibition cells stop growing and dividing as much, and rather, focus on maintaining and repairing to keep the cells and body healthy and ensure a healthy, long life. mTOR can also cause tumor growth but the inhibition can lower the cancer risk. mTOR is also caused primarily by leucine, so with low leucine intake, mTOR is inhibited more.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more vegetables, soy products, green tea, berries, turmeric, and grapes for mTOR inhibition
Evidence	“As discussed earlier, amino acids (of which leucine has the strongest effect) are important signals for mTORC1 activation. Therefore, a significant reduction in leucine intake reduces the activity of the mTORC1 signaling pathway as much as a reduction in total amino acid intake that can be an effective tool for the prevention of several chronic diseases like T2DM, obesity, and cancer As animal-based foods (meat, eggs, milk, and dairy products) contain the highest levels of leucine, the only way to reduce leucine intake is to eliminate or at least strictly reduce the consumption of these food sources and increase the consumption of plant-derived foods that are low in leucine.”
Longevity Impact	With more mTOR inhibition cells stop growing and dividing as much, and rather, focus on maintaining and repairing to keep the cells and body healthy and ensure a healthy, long life. mTOR can also cause tumor growth but the inhibition can lower the cancer risk. mTOR is also caused primarily by leucine, so with low leucine intake, mTOR is inhibited more.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more vegetarian and remove meats from your diet to promote mTOR inhibition effectively
Evidence	“Therefore, a VEG diet result in a lower activation of mTOR-based signaling”
Longevity Impact	With more mTOR inhibition cells stop growing and dividing as much, and rather, focus on maintaining and repairing to keep the cells and body healthy and ensure a healthy, long life. mTOR can also cause tumor growth but the inhibition can lower the cancer risk. mTOR is also caused primarily by leucine, so with low leucine intake, mTOR is inhibited more.
Source	“The Impact of Vegan and Vegetarian Diets on Physical Performance and Molecular Signaling in Skeletal Muscle” by Alexander Pohl et. al

Actionable Recommendation	Start eating more legumes, soy products, nuts, seeds, and whole grains for more plant-based proteins for mTOR inhibition
Evidence	“Ingestion of plant-based proteins induced a lower rise in blood leucine compared to whey, which coincided with a dampened mTORC1 activation.”
Longevity Impact	With more mTOR inhibition cells stop growing and dividing as much, and rather, focus on maintaining and repairing to keep the cells and body healthy and ensure a healthy, long life. mTOR can also cause tumor growth but the inhibition can lower the cancer risk. mTOR is also caused primarily by leucine, so with low leucine intake, mTOR is inhibited more.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more fruits, vegetables, leafy greens, oils, and fats for low leucine for better mTOR inhibition
Evidence	"Plant foods have much less [leucine]: fruits, vegetables, grains, and beans."
Longevity Impact	With more mTOR inhibition cells stop growing and dividing as much, and rather, focus on maintaining and repairing to keep the cells and body healthy and ensure a healthy, long life. mTOR can also cause tumor growth but the inhibition can lower the cancer risk. mTOR is also caused primarily by leucine, so with low leucine intake, mTOR is inhibited more.
Source	Nutrition Facts

4. Telomere Preservation

Actionable Recommendation	Eat more leafy greens, berries, seeds, nuts, and green tea as they provide the antioxidants, fiber, and phytochemicals needed to preserve telomeres the most.
Evidence	“Emerging evidence suggests that dietary patterns rich in plant-based, minimally processed foods may influence telomere dynamics, potentially extending healthspan. This model emphasizes the consumption of whole plant foods, functional bioactives, and the reduction of UPFs to preserve telomere integrity.”

Longevity Impact	Plant-based diets can preserve telomere length, which normally shrinks down with each cell division, but with the length preserved it allows the cell to keep dividing and not experience genomic instability and allow the cell to stay healthy and in turn the body too, increasing lifespan.
Source	[NIH PubMed Central]

Actionable Recommendation	Eat more fresh vegetables, fruits, whole grains, and legumes for telomere preservation
Evidence	"A plant-based dietary pattern rich in healthy plant foods is associated with longer telomeres."
Longevity Impact	Plant-based diets can preserve telomere length, which normally shrinks down with each cell division, but with the length preserved it allows the cell to keep dividing and not experience genomic instability and allow the cell to stay healthy and in turn the body too, increasing lifespan.
Source	[NIH PubMed Central]

Actionable Recommendation	Start eating more antioxidant-rich fruits, vegetables, healthy fats, and fiber and avoid processed foods to preserve telomere length
Evidence	"Diets rich in whole, unprocessed plant foods were associated with longer telomeres."
Longevity Impact	Plant-based diets can preserve telomere length, which normally shrinks down with each cell division, but with the length preserved it allows the cell to keep dividing and not experience genomic instability and allow the cell to stay healthy and in turn the body too, increasing lifespan.
Source	[“Employing Nutrition to Delay Aging: A Plant-Based Telomere-Friendly Dietary Revolution”] by Joanna Polom and Virginia Boccardi

Actionable Recommendation	Eat more carotenoid-rich foods, dark leafy greens, nuts, and seeds for longer telomere length

Evidence	“"Plant-rich dietary patterns" are associated with longer telomere length.”
Longevity Impact	Plant-based diets can preserve telomere length, which normally shrinks down with each cell division, but with the length preserved it allows the cell to keep dividing and not experience genomic instability and allow the cell to stay healthy and in turn the body too, increasing lifespan.
Source	[NIH PubMed Central]()

Actionable Recommendation	Start having a high intake of legumes, vegetables, fruits, grains, nuts, seeds to help lengthen telomeres
Evidence	"Vegetarian diet can prevent or delay telomere shortening."
Longevity Impact	Plant-based diets can preserve telomere length, which normally shrinks down with each cell division, but with the length preserved it allows the cell to keep dividing and not experience genomic instability and allow the cell to stay healthy and in turn the body too, increasing lifespan.
Source	[NIH PubMed Central]()

Actionable Recommendation	Eat more leafy greens, vegetables, berries, nuts, seeds, whole grains, legumes, spices, and teas to increase length of your telomeres
Evidence	"Telomeres... were shown to elongate among men who adopted a low-fat, plant-based diet."
Longevity Impact	Plant-based diets can preserve telomere length, which normally shrinks down with each cell division, but with the length preserved it allows the cell to keep dividing and not experience genomic instability and allow the cell to stay healthy and in turn the body too, increasing lifespan.
Source	[Physicians Committee for Responsible Medicine]()

Okinawan Diet

Main Nutrients
- Purple sweet potato
- Tofu and soy
- Melon
- Leafy greens
- Seaweed
- Seafood
- Jasmine green tea
- Grains
- Pork

Limitations
- Soy or seafood allergies
- Require a high-protein diet
- No access to speciality root vegetables
- Low sodium oriented
- Vegetarian/vegan

Okinawan Diet is primarily related to 4 main cellular mechanisms to increase lifespan:

1. **FOXO3 Activation**

Actionable Recommendation	Eat more purple and orange sweet potatoes, soy foods, turmeric, seafood and marine algae, and bitter melon to activate the FOXO3 gene primarily through caloric restriction, plant-based phytonutrients, and antioxidant-rich whole foods.
Evidence	"A growing consensus favors the hypothesis that this low nutrient intake induces "hormesis", a low-level stress, which triggers the upregulation of stress resistance biological pathways, which include genes encoding for FOXO3."
Longevity Impact	The low nutrient intake that the Okinawa diet supports allows FOXO3 to be activated and enter the cell nucleus to trigger a wide array of genes responsible for DNA repair, antioxidant defense, and cell-cycle arrest. This helps the cells greatly and allows a person to live a longer life with better cellular health. Exceptional longevity in Okinawans has been associated with both the FOXO3 longevity gene and the traditional Okinawan diet, suggesting that genetic and dietary factors may act together to promote healthy aging. So, they necessarily don't have to directly affect one another, but there is correlation and interconnection there.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more okinawa purple sweet potatoes, seaweed, kelp, turmeric, soy products, bitter melon, local greens, mushrooms for FOXO3 activation
Evidence	"The gene FOXO3 is strongly associated with human longevity."
Longevity Impact	The low nutrient intake that the Okinawa diet supports allows FOXO3 to be activated and enter the cell nucleus to trigger a wide array of genes responsible for DNA repair, antioxidant defense, and cell-cycle arrest. This helps the cells greatly and allows a person to live a longer life with better cellular health. Exceptional longevity in Okinawans has been associated with both the FOXO3 longevity gene and the traditional Okinawan diet, suggesting that genetic and dietary factors may act together to promote healthy aging. So, they necessarily don't have to directly affect one another, but there is correlation and interconnection there.
Source	"Exceptional longevity in Okinawa: Demographic trends since 1975" by Michel Poulain and Anne Herm

Actionable Recommendation	Eat more okinawan purple sweet potatoes, soy based foods, and green and jasmine tea for FOXO3 activation
Evidence	"Novel protective effect of the *FOXO3* longevity genotype on mechanisms of cellular aging in Okinawans."
Longevity Impact	The low nutrient intake that the Okinawa diet supports allows FOXO3 to be activated and enter the cell nucleus to trigger a wide array of genes responsible for DNA repair, antioxidant defense, and cell-cycle arrest. This helps the cells greatly and allows a person to live a longer life with better cellular health. Exceptional longevity in Okinawans has been associated with both the FOXO3 longevity gene and the traditional Okinawan diet, suggesting that genetic and dietary factors may act together to promote healthy aging. So, they necessarily don't have to directly affect one another, but there is correlation and interconnection there.
Source	NIH PubMed Central

Actionable Recommendation	Eat more okinawan sweet potatoes, bitter melon, seaweed, soy products, dark leafy greens, spices, and mushroom for FOXO3 activation
Evidence	"A lot of these foods turn on FOXO3, the longevity gene, which slows telomere shortening and reduces inflammation."
Longevity Impact	The low nutrient intake that the Okinawa diet supports allows FOXO3 to be activated and enter the cell nucleus to trigger a wide array of genes responsible for DNA repair, antioxidant defense, and cell-cycle arrest. This helps the cells greatly and allows a person to live a longer life with better cellular health. Exceptional longevity in Okinawans has been associated with both the FOXO3 longevity gene and the traditional Okinawan diet, suggesting that genetic and dietary factors may act together to promote healthy aging. So, they necessarily don't have to directly affect one another, but there is correlation and interconnection there.
Source	National Geographic

Actionable Recommendation	Eat more okinawan sweet potatoes and dark leafy greens for FOXO3 activation
Evidence	"Curcumin, by increasing FoxO3 activity, may protect against oxidant- and lipid-induced damage." (curcumin has a big part in the okinawan diet)
Longevity Impact	The low nutrient intake that the Okinawa diet supports allows FOXO3 to be activated and enter the cell nucleus to trigger a wide array of genes responsible for DNA repair, antioxidant defense, and cell-cycle arrest. This helps the cells greatly and allows a person to live a longer life with better cellular health. Exceptional longevity in Okinawans has been associated with both the FOXO3 longevity gene and the traditional Okinawan diet, suggesting that genetic and dietary factors may act together to promote healthy aging. So, they necessarily don't have to directly affect one another, but there is correlation and interconnection there.
Source	NIH PubMed Central

Actionable Recommendation	Eat more okinawan sweet potatoes, bitter melon, and mushroom for FOXO3 activation
Evidence	"FOXO3 polymorphisms are associated with extreme human longevity."
Longevity Impact	The low nutrient intake that the Okinawa diet supports allows FOXO3 to be activated and enter the cell nucleus to trigger a wide array of genes responsible for DNA repair, antioxidant defense, and cell-cycle arrest. This helps the cells greatly and allows a person to live a longer life with better cellular health. Exceptional longevity in Okinawans has been associated with both the FOXO3 longevity gene and the traditional Okinawan diet, suggesting that genetic and dietary factors may act together to promote healthy aging. So, they necessarily don't have to directly affect one another, but there is correlation and interconnection there.
Source	"Longevity Factor FOXO3: A Key Regulator in Aging-Related Vascular Diseases" by Yan Zhao and You-Shuo Liu

Actionable Recommendation	Start eating more okinawan sweet potatoes, seaweed, and especially turmeric for the curcumin needed for FOXO3 activation
Evidence	"Curcumin induced FOXO3a phosphorylation, FOXO3a nuclear translocation, and increased FOXO3a-mediated gene expression by 2-fold, as measured using a FOXO3a-luciferase reporter gene." (Curcumin has a big part in the Okinawan diet)
Longevity Impact	The low nutrient intake that the Okinawa diet supports allows FOXO3 to be activated and enter the cell nucleus to trigger a wide array of genes responsible for DNA repair, antioxidant defense, and cell-cycle arrest. This helps the cells greatly and allows a person to live a longer life with better cellular health. Exceptional longevity in Okinawans has been associated with both the FOXO3 longevity gene and the traditional Okinawan diet, suggesting that genetic and dietary factors may act together to promote healthy aging. So, they necessarily don't have to directly affect one another, but there is correlation and interconnection there.

Source	“Regulatory effects of curcumin on FOXO3a in human THP-1 monocytes/macrophages” by Jean-Marc Zingg

2. **Suppression of the NF-kB Inflammatory Cascade**

Actionable Recommendation	Eat more purple sweet potatoes, turmeric, soy foods, seaweed, bitter melon, and jasmine and green tea as they give the nutrients to disrupt inflammatory signaling pathways like NF-kB
Evidence	“Curcumin, the bioactive polyphenol in turmeric, is widely used in Okinawan cuisine and herbal teas. It modulates nuclear factor kappa-light-chain-enhancer of activated B cells (NF-κB).”
Longevity Impact	With the Okinawan diet suppressing or modulating the NF-kB inflammatory cascade, it stops chronic inflammation, tissue damage, and tumor growth, allowing a better chance for a longer life. A traditional Okinawan diet has high phytochemicals/antioxidants and leads to reduced oxidative stress which leads to decreased NF-κB activation and as a result, lower inflammation.
Source	Journal of Integrative Dermatology

Actionable Recommendation	Eat more turmeric, seaweed, and jasmine tea to suppress this pathway
Evidence	"The traditional Okinawan diet ... is anti-inflammatory and rich in antioxidants and phytochemicals."
Longevity Impact	With the Okinawan diet suppressing or modulating the NF-kB inflammatory cascade, it stops chronic inflammation, tissue damage, and tumor growth, allowing a better chance for a longer life. A traditional Okinawan diet has high phytochemicals/antioxidants and leads to reduced oxidative stress which leads to decreased NF-κB activation and as a result, lower inflammation.
Source	National Geographic

Actionable Recommendation	Try to include more jasmine tea, bitter melon, and turmeric into your every day eating habits to help stop this pathway

Evidence	"So, 90 plus percent whole food plant-based makes it a highly anti-inflammatory diet, makes it a highly antioxidant diet."
Longevity Impact	With the Okinawan diet suppressing or modulating the NF-kB inflammatory cascade, it stops chronic inflammation, tissue damage, and tumor growth, allowing a better chance for a longer life. A traditional Okinawan diet has high phytochemicals/antioxidants and leads to reduced oxidative stress which leads to decreased NF-κB activation and as a result, lower inflammation.
Source	Nutrition Facts

Actionable Recommendation	Start eating more turmeric, purple sweet potatoes, and orange sweet potatoes to suppress the NF-kB pathway
Evidence	"The Okinawa diet includes foods rich in antioxidants and anti-inflammatory compounds, such as turmeric, which may contribute to reducing inflammation in the body." (curcumin from turmeric is one of the inhibitors of the NF-kB pathway)
Longevity Impact	With the Okinawan diet suppressing or modulating the NF-kB inflammatory cascade, it stops chronic inflammation, tissue damage, and tumor growth, allowing a better chance for a longer life. A traditional Okinawan diet has high phytochemicals/antioxidants and leads to reduced oxidative stress which leads to decreased NF-κB activation and as a result, lower inflammation.
Source	"The Okinawa Diet: Foods and Habits that Boost Longevity" by Kassairis Munoz

Actionable Recommendation	Eat more soy products, seaweed, and turmeric for suppressing the pathway
Evidence	"Several diets have been examined regarding reduction of inflammation and glucose and lipid levels and improvement of cardiovascular health." (reduced inflammation means there is a decreased NF-kB pathway too)
Longevity Impact	With the Okinawan diet suppressing or modulating the NF-kB inflammatory cascade, it stops chronic inflammation, tissue damage, and tumor growth, allowing a better chance for a

	longer life. A traditional Okinawan diet has high phytochemicals/antioxidants and leads to reduced oxidative stress which leads to decreased NF-κB activation and as a result, lower inflammation.
Source	NIH PubMed Central

3. **Mitophagy Stimulation**

Actionable Recommendation	Eat more purple sweet potato, soy products, turmeric, seaweed, and jasmine tea to help stimulate mitophagy more.
Evidence	“Okinawan diet could represent a feasible nutritional approach to reduce the risk of developing age-related cognitive impairment and corresponding disorders via the stimulation of mitophagy.”
Longevity Impact	The Okinawa diet can stimulate mitophagy, the process to remove damaged mitochondria and ensure that enough ATP is produced and toxic waste doesn’t start to get produced. This promotes cell health, and in turn the body health to help live longer. Polyphenols like curcumin, which is a big part of the okinawan diet, also greatly increase mitophagy.
Source	NIH PubMed Central

Actionable Recommendation	Eat more purple and orange sweet potatoes and soy products to increase mitophagy.
Evidence	“Okinawan diet, have been found to exert their protective functions, enhancing the degradation of damaged mitochondria (mitophagy) via the upregulation of mitophagy.”
Longevity Impact	The Okinawa diet can stimulate mitophagy, the process to remove damaged mitochondria and ensure that enough ATP is produced and toxic waste doesn’t start to get produced. This promotes cell health, and in turn the body health to help live longer. Polyphenols like curcumin, which is a big part of the okinawan diet, also greatly increase mitophagy.
Source	“The Interaction of Diet and Mitochondrial Dysfunction in Aging and Cognition” by Aleksandra Kaliszewska et. al

Actionable Recommendation	Try to eat more seaweed and turmeric to stimulate mitophagy in your body
Evidence	“A range of phytochemicals and zoochemicals are described as promoting mitophagy…Many of these phytochemicals are found in Mediterranean and Okinawan diets.”
Longevity Impact	The Okinawa diet can stimulate mitophagy, the process to remove damaged mitochondria and ensure that enough ATP is produced and toxic waste doesn’t start to get produced. This promotes cell health, and in turn the body health to help live longer. Polyphenols like curcumin, which is a big part of the okinawan diet, also greatly increase mitophagy.
Source	“The Critical Role of Autophagy and Phagocytosis in the Aging Brain” by Stephen C. Bondy and Meixia Wu

Actionable Recommendation	Start eating more jasmine and green tea and soy products to increase mitophagy in the body
Evidence	“Polyphenols in modulating mitophagy.”
Longevity Impact	The Okinawa diet can stimulate mitophagy, the process to remove damaged mitochondria and ensure that enough ATP is produced and toxic waste doesn’t start to get produced. This promotes cell health, and in turn the body health to help live longer. Polyphenols like curcumin, which is a big part of the okinawan diet, also greatly increase mitophagy.
Source	NIH PubMed Central

Actionable Recommendation	Eat more bitter melon, mushrooms, and turmeric to stimulate mitophagy more
Evidence	“Curcumin is another significant polyphenol that regulates mitophagy.”
Longevity Impact	The Okinawa diet can stimulate mitophagy, the process to remove damaged mitochondria and ensure that enough ATP is produced and toxic waste doesn’t start to get produced. This promotes cell health, and in turn the body health to help live longer. Polyphenols like curcumin, which is a big part of the okinawan diet, also greatly increase mitophagy.

Source	[“The role of polyphenols in modulating mitophagy: Implications for therapeutic interventions”]() by Xinyu Lin et. al

Actionable Recommendation	Start eating more soy products, turmeric, and bitter melon to increase mitophagy
Evidence	“Polyphenols have recently been demonstrated to protect mitochondrial health by facilitating mitophagy.”
Longevity Impact	The Okinawa diet can stimulate mitophagy, the process to remove damaged mitochondria and ensure that enough ATP is produced and toxic waste doesn’t start to get produced. This promotes cell health, and in turn the body health to help live longer. Polyphenols like curcumin, which is a big part of the okinawan diet, also greatly increase mitophagy.
Source	NIH PubMed Central

4. **mTOR Inhibition**

Actionable Recommendation	Eat more purple and orange sweet potatoes, soy products, bitter melon, turmeric, green and jasmine tea, and seaweed as mTOR inhibition happens mainly through caloric restriction.
Evidence	“Decreased activity of mTOR and insulin/IGF-1/GH.”
Longevity Impact	The Okinawa diet decreases activity of mTOR, which allows it to focus less on cell growth and more on cell cleanup, keeping a healthy body, and allowing a longer life. Low protein and low fat intake also decreases mTOR activity.
Source	NIH PubMed Central

Actionable Recommendation	Eat more purple sweet potato, soy products, and turmeric to stop the mTOR pathway
Evidence	“Why is the Okinawan diet conducive to longevity?...Moderate protein intake…This ensures low activation of growth-promoting signaling pathways (e.g., mTOR).”
Longevity Impact	The Okinawa diet decreases activity of mTOR, which allows it to focus less on cell growth and more on cell cleanup, keeping

	a healthy body, and allowing a longer life. Low protein and low fat intake also decreases mTOR activity.
Source	Sanotact

Actionable Recommendation	Start eating more green tea, seaweed, and bitter melon to inhibit mTOR
Evidence	"Traditional Okinawan diets were historically very low in protein."
Longevity Impact	The Okinawa diet decreases activity of mTOR, which allows it to focus less on cell growth and more on cell cleanup, keeping a healthy body, and allowing a longer life. Low protein and low fat intake also decreases mTOR activity.
Source	"Low-Protein Diets for Longevity: Benefits, Risks, and Scientific Evidence" by Hugo Francisco de Souza

Actionable Recommendation	Try to eat more wild greens, radish, and bamboo shoots to stop activating this pathway as much
Evidence	"The traditional Okinawa diet is low in calories and fat while high in carbs."
Longevity Impact	The Okinawa diet decreases activity of mTOR, which allows it to focus less on cell growth and more on cell cleanup, keeping a healthy body, and allowing a longer life. Low protein and low fat intake also decreases mTOR activity.
Source	"What Is the Okinawa Diet? Foods, Longevity, and More" by Ansley Hill

Actionable Recommendation	Eat more seaweed and soy products like tofu to stop this pathway
Evidence	"The traditional Okinawan diet is low in fat (27% of energy)."
Longevity Impact	The Okinawa diet decreases activity of mTOR, which allows it to focus less on cell growth and more on cell cleanup, keeping a healthy body, and allowing a longer life. Low protein and low fat intake also decreases mTOR activity.

Source	[The University of Sydney]

Actionable Recommendation	Start eating more turmeric, soy products, and bitter melon to decrease the activity of mTOR
Evidence	“The traditional Okinawan diet is the lowest in fat intake, particularly in terms of saturated fat.”
Longevity Impact	The Okinawa diet decreases activity of mTOR, which allows it to focus less on cell growth and more on cell cleanup, keeping a healthy body, and allowing a longer life. Low protein and low fat intake also decreases mTOR activity.
Source	[Research Gate]

DASH Diet

Main Nutrients
- Whole grains
- Vegetables
- Fruits
- Lowfat/Fat-free dairy
- Lean Meats
- Poultry
- Fish
- Nuts
- Seeds
- Legumes
- Fats and oils

Limitations
- Expensive
- CKD conditions
- Sensitive to sudden increase in fiber
- Sodium deficiency
- People taking ACE inhibitors

DASH Diet is primarily related to 5 main cellular mechanisms to increase lifespan:

1. **eNOS Optimization**

Actionable Recommendation	Eat more leafy greens, root vegetables, berries, nuts, seeds, fatty fish, and legumes to optimize eNOS function.

Evidence	“Several components of the DASH diet may be partly responsible for its salutary effects and influence on endothelial function.”
Longevity Impact	The DASH diet helps support endothelial tissue which is supported by the enzyme eNOS that produces nitric oxide to support this tissue. With the DASH diet improving endothelial function, it means that eNOS is being optimized and this enzyme is functioning well.
Source	NIH PubMed Central

Actionable Recommendation	Eat more leafy greens, root vegetables, and berries for eNOS optimization
Evidence	"The DASH intervention proved more effective than routine care in... improvement in endothelial function."
Longevity Impact	The DASH diet helps support endothelial tissue which is supported by the enzyme eNOS that produces nitric oxide to support this tissue. With the DASH diet improving endothelial function, it means that eNOS is being optimized and this enzyme is functioning well.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more nuts, seeds, legumes, and fish to optimize the eNOS enzyme
Evidence	"Arginine, citrulline, cocoa polyphenols, folate, vitamin C, and magnesium can help protect and maintain NO synthesis." (all of these nutrients are in the DASH diet)
Longevity Impact	The DASH diet helps support endothelial tissue which is supported by the enzyme eNOS that produces nitric oxide to support this tissue. With the DASH diet improving endothelial function, it means that eNOS is being optimized and this enzyme is functioning well.
Source	Metagenics Institute

Actionable Recommendation	Try to eat more plain nonfat greek yogurt, fat-free milk, and whole grains for eNOS optimization

Evidence	"The DASH diet increased serum nitrite concentrations compared with the control diet."
Longevity Impact	The DASH diet helps support endothelial tissue which is supported by the enzyme eNOS that produces nitric oxide to support this tissue. With the DASH diet improving endothelial function, it means that eNOS is being optimized and this enzyme is functioning well.
Source	[Why Don't Physicians Follow Clinical Practice Guidelines?]() by Michael D. Cabana et. al

Actionable Recommendation	Eat more pomegranates and extra virgin olive oil to optimize the eNOS enzyme
Evidence	"Potassium supplementation improved endothelial function." (DASH diet is high in potassium)
Longevity Impact	The DASH diet helps support endothelial tissue which is supported by the enzyme eNOS that produces nitric oxide to support this tissue. With the DASH diet improving endothelial function, it means that eNOS is being optimized and this enzyme is functioning well.
Source	NIH PubMed Central

Actionable Recommendation	Start eating more baked potatoes, broccoli, cauliflower, and other vegetables for eNOS optimization
Evidence	“(DASH) diet was associated with changes in peripheral vascular function: endothelial function, assessed by flow-mediated vasodilatation (FMD).”
Longevity Impact	The DASH diet helps support endothelial tissue which is supported by the enzyme eNOS that produces nitric oxide to support this tissue. With the DASH diet improving endothelial function, it means that eNOS is being optimized and this enzyme is functioning well.
Source	NIH PubMed Central

2. Optimization of the Na^+/K^+-ATPase Pump

Actionable Recommendation	Eat more potassium-rich fruits and vegetables, nuts for magnesium sources, and whole grains to optimize this pump
Evidence	“A potassium-rich diet…was associated with lower blood pressure and higher vascular Na,K-ATPase activity.”
Longevity Impact	The DASH diet is a potassium-rich diet that can increase the Na+/k+ ATPase Pump activity which can maintain membrane potential and control cellular volume, which can keep the cell stable and healthy and ultimately lead to a longer life with healthier cells.
Source	NIH PubMed Central

Actionable Recommendation	Eat more root vegetables, leafy greens, fruits, and legumes to get high-potassium for the pump
Evidence	“Use DASH Diet To Increase Potassium.”
Longevity Impact	The DASH diet is a potassium-rich diet that can increase the Na+/k+ ATPase Pump activity which can maintain membrane potential and control cellular volume, which can keep the cell stable and healthy and ultimately lead to a longer life with healthier cells.
Source	UCLA Health

Actionable Recommendation	Start eating more low-fat dairy, nuts, seeds, and omega-3 fish to activate the enzyme pump
Evidence	"Adherence to the DASH diet enhances potassium intake.”
Longevity Impact	The DASH diet is a potassium-rich diet that can increase the Na+/k+ ATPase Pump activity which can maintain membrane potential and control cellular volume, which can keep the cell stable and healthy and ultimately lead to a longer life with healthier cells.
Source	NIH PubMed Central

Actionable Recommendation	Try eating more whole grains, fruits, vegetables, and lean proteins to activate the pump

Evidence	"The DASH diet helps people lower salt... The diet is also rich in nutrients that help lower blood pressure. These include potassium."
Longevity Impact	The DASH diet is a potassium-rich diet that can increase the Na+/k+ ATPase Pump activity which can maintain membrane potential and control cellular volume, which can keep the cell stable and healthy and ultimately lead to a longer life with healthier cells.
Source	Mayo Clinic

Actionable Recommendation	Eat more seeds, nuts, legumes, and vegetables to optimize the pump
Evidence	"The DASH eating plan includes good sources of potassium."
Longevity Impact	The DASH diet is a potassium-rich diet that can increase the Na+/k+ ATPase Pump activity which can maintain membrane potential and control cellular volume, which can keep the cell stable and healthy and ultimately lead to a longer life with healthier cells.
Source	American Heart Association

3. **Suppression of the Angiotensin II / AT1R Signaling Pathway**

Actionable Recommendation	Eat more berries, grapes, green and black tea, spinach and dark leafy greens, beans, lentils, and nuts to get more Mg and inhibit this pathway.
Evidence	“Mg inhibits the renin–angiotensin–aldosterone system, by counteracting angiotensin II and inhibiting aldosterone production.”
Longevity Impact	The DASH diet gives a lot of Mg, which can inhibit the Angiotensin II pathway and stop a cascade that triggers cellular inflammation and fluid retention, keeping the cells and body healthy to live longer. The DASH diet however interacts with and modifies the renin–angiotensin–aldosterone system (RAAS) first, the pathway that produces and signals through Angiotensin II and AT1 receptors. ACE is another enzyme that also produces Angiotensin, so that can be stopped by the DASH diet too.

Source	[Cambridge]()

Actionable Recommendation	Eat more berries and potassium-rich fruits and veggies to suppress this pathway
Evidence	"The DASH diet interacts with the RAAS resulting in vascular and hormonal responses."
Longevity Impact	The DASH diet gives a lot of Mg, which can inhibit the Angiotensin II pathway and stop a cascade that triggers cellular inflammation and fluid retention, keeping the cells and body healthy to live longer. The DASH diet however interacts with and modifies the renin–angiotensin–aldosterone system (RAAS) first, the pathway that produces and signals through Angiotensin II and AT1 receptors. ACE is another enzyme that also produces Angiotensin, so that can be stopped by the DASH diet too.
Source	[NIH PubMed Central]()

Actionable Recommendation	Start eating more cocoa beans, dark chocolate, and green tea to stop this pathway
Evidence	"DASH diet lowers BP through effects on vascular function and sodium excretion, mediated through the effects on RAAS."
Longevity Impact	The DASH diet gives a lot of Mg, which can inhibit the Angiotensin II pathway and stop a cascade that triggers cellular inflammation and fluid retention, keeping the cells and body healthy to live longer. The DASH diet however interacts with and modifies the renin–angiotensin–aldosterone system (RAAS) first, the pathway that produces and signals through Angiotensin II and AT1 receptors. ACE is another enzyme that also produces Angiotensin, so that can be stopped by the DASH diet too.
Source	[Clinical Trials]()

Actionable Recommendation	Try eating more garlic, olive oil, and berries to suppress this pathway and keep the cells healthy

Evidence	"The DASH diet causes natriuretic and diuretic effects, increases the amount of physiological inhibitory effects of the angiotensin converter enzyme."
Longevity Impact	The DASH diet gives a lot of Mg, which can inhibit the Angiotensin II pathway and stop a cascade that triggers cellular inflammation and fluid retention, keeping the cells and body healthy to live longer. The DASH diet however interacts with and modifies the renin–angiotensin–aldosterone system (RAAS) first, the pathway that produces and signals through Angiotensin II and AT1 receptors. ACE is another enzyme that also produces Angiotensin, so that can be stopped by the DASH diet too.
Source	Clinical Hypertension

Actionable Recommendation	Eat more leafy greens, vegetables, nuts, seeds, and legumes to stop this pathway from happening as much
Evidence	"Postulated mechanisms for the BP-lowering effect of the DASH diet include effects on...renin-angiotensin-aldosterone system (RAAS)."
Longevity Impact	The DASH diet gives a lot of Mg, which can inhibit the Angiotensin II pathway and stop a cascade that triggers cellular inflammation and fluid retention, keeping the cells and body healthy to live longer. The DASH diet however interacts with and modifies the renin–angiotensin–aldosterone system (RAAS) first, the pathway that produces and signals through Angiotensin II and AT1 receptors. ACE is another enzyme that also produces Angiotensin, so that can be stopped by the DASH diet too.
Source	Cambridge

4. **Glycemic Stability**

Actionable Recommendation	Eat more whole grains, legumes, non-starchy vegetables, lean proteins, and low-fat dairy to increase glycemic stability.
Evidence	“Participants eating the DASH4D diet (DASH diet but more diabetes-friendly) had blood sugar levels that were on average 11 mg/dL lower than when eating the standard diet and stayed

	in the optimal blood glucose range for an extra 75 minutes a day."
Longevity Impact	With more glycemic stability from the DASH diet, the blood sugar spikes that usually cause sugar molecules to bind to proteins and then to (RAGE) receptors that trigger a cascade which stiffens cells and tissues doesn't happen, promoting cellular health and allowing the body to stay healthy for longer to live longer. DASH diet also affects insulin sensitivity, and a high insulin sensitivity or action has better glycemic stability.
Source	John Hopkins

Actionable Recommendation	Start eating more non-starchy vegetables and whole grains to help your glycemic stability
Evidence	"The DASH diet significantly reduces the risk of type 2 diabetes by optimizing blood glucose homeostasis."
Longevity Impact	With more glycemic stability from the DASH diet, the blood sugar spikes that usually cause sugar molecules to bind to proteins and then to (RAGE) receptors that trigger a cascade which stiffens cells and tissues doesn't happen, promoting cellular health and allowing the body to stay healthy for longer to live longer. DASH diet also affects insulin sensitivity, and a high insulin sensitivity or action has better glycemic stability.
Source	"The DASH diet in diabetes related complications or comorbidities: an unexpected friend" by Kai Liu et. al

Actionable Recommendation	Try to eat more legumes, nuts, and healthy fats for your glycemic stability
Evidence	"The DASH dietary pattern may lead to an improvement in insulin sensitivity independent of weight loss."
Longevity Impact	With more glycemic stability from the DASH diet, the blood sugar spikes that usually cause sugar molecules to bind to proteins and then to (RAGE) receptors that trigger a cascade which stiffens cells and tissues doesn't happen, promoting cellular health and allowing the body to stay healthy for longer to live longer. DASH diet also affects insulin sensitivity, and a high insulin sensitivity or action has better glycemic stability.

Source	NIH PubMed Central

Actionable Recommendation	Eat more lean protein, fish, and low-fat dairy for better glycemic stability
Evidence	"Including the DASH dietary pattern in a comprehensive intervention for blood pressure control enhanced insulin action."
Longevity Impact	With more glycemic stability from the DASH diet, the blood sugar spikes that usually cause sugar molecules to bind to proteins and then to (RAGE) receptors that trigger a cascade which stiffens cells and tissues doesn't happen, promoting cellular health and allowing the body to stay healthy for longer to live longer. DASH diet also affects insulin sensitivity, and a high insulin sensitivity or action has better glycemic stability.
Source	"DASH Diet Improves Insulin Sensitivity as Well as Hypertension" by Laurie Barclay et. al

Actionable Recommendation	Eat more fiber-rich vegetables and low-glycemic fruits like berries or apples or pears to have a better glycemic stability in your body
Evidence	"Greater adherence to the DASH diet was associated with lower fasting insulin concentrations."
Longevity Impact	With more glycemic stability from the DASH diet, the blood sugar spikes that usually cause sugar molecules to bind to proteins and then to (RAGE) receptors that trigger a cascade which stiffens cells and tissues doesn't happen, promoting cellular health and allowing the body to stay healthy for longer to live longer. DASH diet also affects insulin sensitivity, and a high insulin sensitivity or action has better glycemic stability.
Source	Nutrients Journal

5. Enhancement of Endogenous Antioxidant Defenses

Actionable Recommendation	Eating more fruits, vegetables, nuts, seeds, legumes, whole grains, and lean meats can help as they are antioxidant dense foods that can trigger the antioxidant defense system.

Evidence	“Our results demonstrated that a DASH diet could significantly increase GSH and decrease MDA levels.”
Longevity Impact	GSH is a primary component of the antioxidant defense system which stops the body from producing too many antioxidants which can lead to premature cell death. The DASH diet produces a lot of it to help the body. So, with less cell death, the body stays healthy and a person can live for much longer. The endogenous antioxidant defenses neutralizes free radicals, repairs existing damage, and prevents future oxidative stress. So, if there is less oxidative stress that means more endogenous antioxidant defenses are involved.
Source	NIH PubMed Central

Actionable Recommendation	Eat more vegetables and fruits to enhance the endogenous antioxidant defenses
Evidence	"The DASH-CD raises antioxidant capacity."
Longevity Impact	GSH is a primary component of the antioxidant defense system which stops the body from producing too many antioxidants which can lead to premature cell death. The DASH diet produces a lot of it to help the body. So, with less cell death, the body stays healthy and a person can live for much longer. So, if there is less oxidative stress that means more endogenous antioxidant defenses are involved.
Source	NIH PubMed Central

Actionable Recommendation	Try eating more nuts, seeds, and legumes to enhance the defenses
Evidence	“The DASH-CD raises antioxidant capacity, lowers BP, and reduces oxidative stress.”
Longevity Impact	GSH is a primary component of the antioxidant defense system which stops the body from producing too many antioxidants which can lead to premature cell death. The DASH diet produces a lot of it to help the body. So, with less cell death, the body stays healthy and a person can live for

	much longer. So, if there is less oxidative stress that means more endogenous antioxidant defenses are involved.
Source	American Heart Association Journals

Actionable Recommendation	Start eating more whole grains and healthy fats to enhance the endogenous antioxidant defenses
Evidence	“The results suggest that LS-DASH decreases oxidative stress.”
Longevity Impact	GSH is a primary component of the antioxidant defense system which stops the body from producing too many antioxidants which can lead to premature cell death. The DASH diet produces a lot of it to help the body. So, with less cell death, the body stays healthy and a person can live for much longer. So, if there is less oxidative stress that means more endogenous antioxidant defenses are involved.
Source	NIH PubMed Central

Actionable Recommendation	Eat more allium vegetables and Vitamin-C rich fruits like kiwi and papaya to enhance the endogenous antioxidant defenses
Evidence	“DASH dietary pattern may also restore the activity of antioxidant enzymes.”
Longevity Impact	GSH is a primary component of the antioxidant defense system which stops the body from producing too many antioxidants which can lead to premature cell death. The DASH diet produces a lot of it to help the body. So, with less cell death, the body stays healthy and a person can live for much longer. So, if there is less oxidative stress that means more endogenous antioxidant defenses are involved.
Source	“The effect of the dietary approaches to stop hypertension diet on total antioxidant capacity, superoxide dismutase, catalase, and body composition in patients with non-alcoholic fatty liver disease: a randomized controlled trial” by Zahra Zare et. al

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Glossary

1. **Telomere**
   A protective cap made of repetitive DNA sequences found at the ends of our chromosomes, because when our cell divides a small piece of the chromosome's end is left uncopied and lost. Telomeres take this loss, ensuring that essential, life-sustaining genetic codes remain untouched.

2. **Membrane**
   A thin, flexible barrier that selectively allows some substances to pass through while blocking others. It can surround multiple things, such as cells, organs, and organelles.

3. **Mitochondria**
   A specialized structure found inside most complex cells that acts as the cell's power plant, turning nutrients into usable chemical energy.

4. **ATP**
   The usable chemical energy that the mitochondria produces by breaking down glucose. All organisms use this for basic functions.

5. **Gut microbiota**
   The vast community of trillions of microorganisms—including bacteria, viruses, fungi, and archaea—that live inside your digestive tract, primarily in the large intestine.

6. **Autophagy**
   The body's cellular recycling process.Your cells deliberately break down their own damaged, old, or dysfunctional parts (like broken proteins or worn-out organelles), recycle the pieces, and reuse them for energy or cellular renovation.

7. **AMPK**
   The body's master cellular energy sensor. It acts as a metabolic thermostat, constantly monitoring the amount of energy available inside your cells and shifting your metabolism to keep you alive during times of physical stress. When energy is low, AMPK switches off processes that consume energy and switches on processes that create energy.

8. **mTOR**
   The body's master cellular growth sensor. It acts as the direct counterpart and antagonist to AMPK. While AMPK is the energy sensor that triggers cellular repair during scarcity, mTOR is the biological accelerator that triggers construction, cell division, and tissue growth during times of nutrient abundance.

9. **FOXO3**
   A master gene and protein that acts as the body's ultimate longevity switch. When activated, FOXO3 is the high-level executive that enters the cell’s command center (the nucleus) to physically rewrite gene expression. It instructs your cells to aggressively

repair themselves, resist environmental stress, and directly delay the biological processes of aging.

10. **NF-KB Cascade**
    The body's primary master switch for inflammation and immune responses. When your body detects an injury, a viral attack, or bacterial toxins, it activates the NF-κB cascade. This system forcefully rewrites gene expression to trigger acute inflammation, deploy white blood cells, and keep damaged cells alive to fight off the threat.

11. **Mitophagy**
    The selective, targeted destruction and recycling of damaged or dysfunctional mitochondria through autophagy.While general autophagy digests any generic cellular waste, mitophagy is a specialized quality-control system. It identifies worn-out cellular power plants and safely shreds them before they can leak toxic reactive oxygen species (free radicals) and kill the cell

12. **eNOS**
    eNOS (Endothelial Nitric Oxide Synthase) is a specialized enzyme responsible for producing nitric oxide (NO), a critical signaling gas that controls blood flow, blood pressure, and vascular health.

13. **Angiotensin II Pathway**
    A hormone cascade that plays a central role in regulating blood pressure, fluid volume, and systemic vascular resistance. While it is crucial for surviving short-term physical trauma, its chronic activation represents a powerful counter-current to the body's repair machinery. Over time, an overactive Angiotensin II pathway accelerates aging and tissue damage by forcefully overriding the protective mechanisms of AMPK, FOXO3, and eNOS.

14. **Glycemic Stability**
    The body's ability to maintain blood glucose (sugar) levels within a narrow, healthy range.

15. **Enhancement of Endogenous Antioxidant Defenses**
    Upregulation of the body's internal, self-manufactured antioxidant enzymes. Unlike dietary antioxidants (such as Vitamin C or E) which act as "one-to-one" neutralizers, endogenous antioxidants are reusable catalytic machines. A single internal enzyme can neutralize millions of destructive free radicals every second, making them the body's primary defense against cellular aging.